and

$$\dim V = 2n.$$

Since $V_3 \subseteq V_1 \oplus V_2$, we can write every element of V_3 in the form

$$x \in V_3, \quad x = (x_1, x_2),\ x_1 \in V_1,\ x_2 \in V_2.$$

We then can define the projections

$$B_1 : V_3 \to V_1, \quad (x_1, x_2) \mapsto x_1,$$

$$B_2 : V_3 \to V_2, \quad (x_1, x_2) \mapsto x_2.$$

Since $V_3 \cap V_1 = 0$ and $V_3 \cap V_2 = 0$, these maps have to be injective and therefore are isomorphisms. We then define the isomorphism

$$A = B_2 \circ B_1^{-1} : V_1 \to V_2.$$

Let $e_1, \ldots, e_n$ be a basis for V_1. Then we get

$$\begin{aligned} V_1 &= k\,e_1 \oplus k\,e_2 \oplus \cdots \oplus k\,e_n, \\ V_2 &= k\,Ae_1 \oplus k\,Ae_2 \oplus \cdots \oplus k\,Ae_n, \\ V_3 &= k\,(e_1 + Ae_1) \oplus k\,(e_2 + Ae_2) \oplus \cdots \oplus k\,(e_n + Ae_n). \end{aligned}$$

So we can think of V_3 as the graph of an isomorphism $A : V_1 \to V_2$. From this we obtain the decomposition

$$\begin{tikzcd} \bullet_{V_1} \arrow[r, hook] & \bullet_{V} & \bullet_{V_3} \arrow[l, hook'] \\ & \bullet_{V_2} \arrow[u, hook] & \end{tikzcd} = \bigoplus_{j=1}^{n} \begin{tikzcd} \bullet_{k(1,0)} \arrow[r, hook] & \bullet_{k^2} & \bullet_{k(1,1)} \arrow[l, hook'] \\ & \bullet_{k(0,1)} \arrow[u, hook] & \end{tikzcd}.$$

These correspond to the indecomposable object

$$\begin{tikzcd} \bullet_{1} \arrow[r] & \bullet_{2} & \bullet_{1} \arrow[l] \\ & \bullet_{1} \arrow[u] & \end{tikzcd}.$$

Thus the quiver D_4 with the selected orientation has 12 indecomposable objects. If one were to explicitly decompose representations for the other possible orientations, one would also find 12 indecomposable objects.